UNCOVER: Ultra-deep NIRCam and NIRSpec Observations Before the Epoch of Reionization

Scientific Category: Galaxies

Scientific Keywords: Galaxy Evolution, Galaxy Formation, High-Redshift Galaxies

Instruments: NIRSPEC, NIRISS, NIRCAM

Proposal Size: MEDIUM

Exclusive Access Period: 0 months

Allocation Information (in hours):

Science Time: 50.2

Coordinated Parallel Time: 49.8

Charged Time: 68.2

Treasury: Yes

Abstract

We propose an efficient public Treasury program that immediately establishes a NIRCam imaging deep field and ultra-deep low-resolution NIRSpec/PRISM follow-up spectroscopy in the gravitational lensing cluster Frontier Field Abell 2744. Assisted by strong lensing, these observations reach 1-2 magnitudes fainter than even the deepest ERS & GTO programs. Such depths are essential to achieve two core science goals of JWST: finding First Light galaxies during the Dark Ages at z>10 and studying the ultra-low luminosity galaxies at later times that were responsible for reionization. Offering the community early access to deep imaging of 4000 z>6 galaxies and spectroscopy of 500 galaxies ensures that this envisioned flagship science is guaranteed early in the mission, establishes from the start a vibrant and diverse user base for the observatory, and optimizes the efficiency of JWST by providing targets for higher resolution spectroscopic follow up in subsequent cycles. In support of this, we included imaging parallels to enhance the deep imaging legacy on and around the cluster. Beyond the immediate science goals, these data will support a broad array of legacy science including stellar mass complete studies to z=10, the role of dust obscuration at high redshift, and the various pathways of quenching star formation. Our experienced team commits to rapidly releasing the imaging to the public before the Cycle 2 deadline followed by the delivery of a joint photometric and spectroscopic database.

Scientific Justification

"In 1995, Astronomer Bob Williams wanted to point the Hubble Space Telescope at a patch of sky filled with absolutely nothing remarkable. For 100 hours. It was a terrible idea, his colleagues told him, and a waste of valuable telescope time. People would kill for that amount of time with the sharpest tool in the shed, they said, and besides — no way would the distant galaxies Williams hoped to see be bright enough for Hubble to [uncover]..."

– Nadia Drake

First light and Reionization: The *James Webb Space Telescope* was specifically designed to find the first galaxies deep in the Dark Ages (z=10−20) and directly study the faint galaxies responsible for reionizing the universe a few hundred million years later. Achieving these core science goals requires deep imaging to locate the galaxies and ultra-deep spectroscopy at 1−5μm to confirm their redshifts. The Early Release Science programs are too shallow for First Light science. Even the deepest planned NIRCam & NIRSpec GTO program (PI: Eisenstein) is scarcely sensitive enough, only covers one field and will have no initial public access. In the spirit of the original Hubble Deep Field, we designed the Ultra-deep Nirspec and nirCam ObserVations before the Epoch of Reionization (UNCOVER) Treasury program.

The key strengths are to push **ultra-deep** *with* **NIRCam imaging** *and* **NIRSpec spectroscopy** *in a* **low background field** *assisted by* **strong gravitational lensing** *and to make these data* **immediately public.**

UNCOVER Synopsis: The program consists of two coordinated components: 1) deep NIRCam pre-imaging mosaic in 8 filters for 5 hours per band and in the same cycle 2) ultra-deep 20 hour NIRSpec/PRISM low-resolution follow up of the NIRCam-detected high-redshift galaxies. The observations are located in the powerful lensing cluster Abell 2744, which features low infrared background and a high magnification area that is well matched to the NIRCam field of view. UNCOVER by itself will **double the lensed imaging area of the entire Hubble Frontier Field program and its parallels with ∼1.5−3 magnitudes deeper photometry at 1 − 5μm.** The NIRSpec/PRISM is a perfect match for immediate spectroscopic exploration of the NIRCam-discovered galaxies. The outstanding wavelength coverage 0.6−5.3μm and continuum sensitivity not only guarantees the ability to measure the redshifts of "Dark Age" galaxies at z>10 and faint reionization-era galaxies, regardless of the presence of emission lines, but they also support the broadest possible range of legacy science investigations. As with the Hubble Deep Field, these imaging and spectra are certain to receive intense interest from our community, providing an iconic dataset for scientific firsts and a wealth of sources both for immediate study and as targets for follow-up with JWST and ALMA. **The key distinguishing feature of UNCOVER is the gravitational lensing boost, allowing observations to extend fainter intrinsically than any existing HST or planned JWST ERS or GTO program**, enabling unique science and safeguarding their long-term legacy value (Fig. 1).

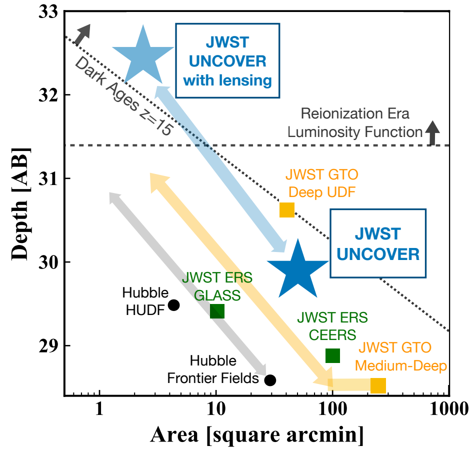

Figure 1: The UNCOVER program in the context of existing HST (black) deep fields and planned JWST ERS (green) and GTO (orange) programs. The arrows indicate the cosmic zoom effect of gravitational lensing: increasing depth but reducing area. By combining deep observations with strong lensing from a foreground cluster, only UNCOVER will explore the necessary depths catch the first galaxies and to probe the faint end of the reionization era luminosity function. The $\sim$50 sq. arcmin imaging component of UNCOVER will jumpstart community analysis of JWST deep fields early in the mission, enhanced further by rapid follow up low-resolution spectroscopy without proprietary time.

Why now? The value of these ultra-deep imaging and spectra is maximal if taken in Cycle 1. The outlined plan ensures that the envisioned flagship science is done early in the mission, establishes a vibrant and diverse user base for the observatory from day one, optimizes the efficiency of JWST by providing targets for higher spatial/spectral resolution follow up in subsequent cycles, and opens up new parameter space for unexpected discoveries quickly.

Key Science enabled by JWST and UNCOVER

The proposed NIRCam imaging will improve over HST/WFC3, ground-based K-band, and Spitzer sensitivity by a factor $6-20\times$ overnight, while ultra-sensitive continuum spectroscopy beyond λ>2.5μm remains impossible from the ground due to atmospheric absorption and high thermal background. These improvements are absolute game changers in studies of galaxy formation. UNCOVER combines both ultra-deep imaging and spectroscopy, assisted by gravitational lensing to achieve depths corresponding to >32AB magnitudes, to tackle key science topics. Such depths would be prohibitively expensive to achieve in blank fields, even with JWST (corresponding to >1Ms integrations per filter).

I. First Light

A lasting legacy of Hubble and Spitzer is the discovery and characterization of galaxies to z~11, looking back 97% of the time to the Big Bang (e.g., Oesch+2016). Beyond lies the realm of the Dark Ages (10<z<20), the period after Recombination when the first stars and galaxies formed. Our understanding of galaxy formation during this period is completely theoretical and varies wildly between models (Fig. 2). Some suggest galaxy evolution should extend beyond z>14 (e.g., Dayal+2014, Hutter+2020), others predict a steep decline at z>8, in apparent agreement with HST observations (e.g., Oesch+2018). In addition, almost nothing is known about the earliest star formation histories. Few galaxies have been spectroscopically confirmed at z>9 and their properties do not tell a clear story: some galaxies appear extremely young (5 Myrs old, e.g., Endsley+2020), while others are thought to have formed 100s of Myrs earlier (z=12−20, Hashimoto+2018).

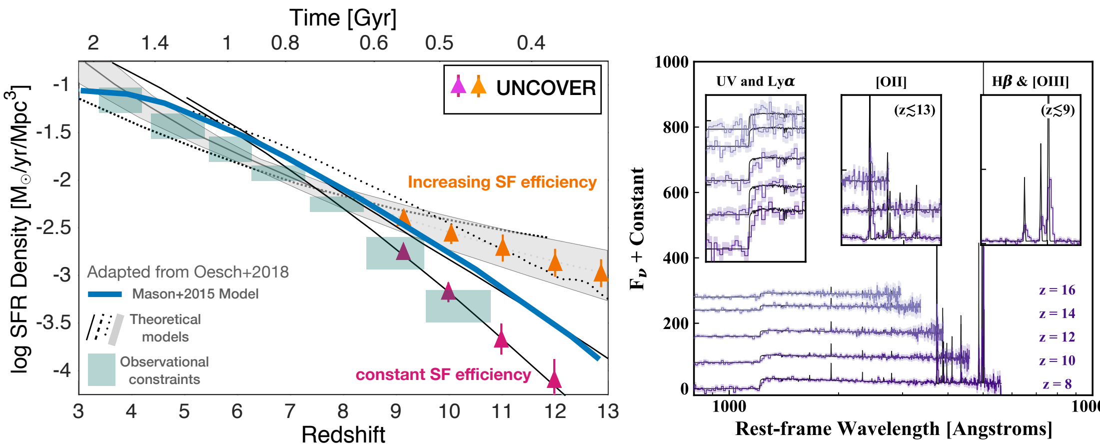

Figure 2: Left: The integrated star formation history of the Universe is barely constrained beyond $z \gtrsim 8$; UNCOVER will differentiate amongst theoretical models that adopt different prescriptions for star formation efficiency. Right: in 20 hours, UNCOVER will provide unprecedented continuum spectra to 28−29AB magnitude at $z>10$. The Balmer break will constrain ages and emission lines reveal key physical properties

What UNCOVER enables: Our ultra-deep imaging extends photometric studies robustly beyond $z>10$. Here the NIRCam filters track the redshifted position of the strong Lyman-break and we can select them with traditional dropout selection and photometric redshifts. Theoretical model predictions of the number of galaxies vary by an order of magnitude (see Fig. 2), depending on the assumed efficiency of star formation in dark matter halos. Broadly speaking, increased efficiency towards earlier times means slower evolution. Using the "middle-of-the-road" Mason+(2015) model (see Technical Justification) we expected to find $\sim$150 galaxies at z=9−11, another 40 at z=11−13, and $\sim$6 at z=13−15, of which 70% are expected over highly lensed areas. UNCOVER will provide definitive constraints on these models and will help calibrate our predictions for $z>15$.

UNCOVER will confirm the redshifts of "Dark Age" galaxies immediately in Cycle 1. The proposed $R\sim100$ PRISM spectra provide unbiased spectroscopic continuum redshifts of any JWST-selected galaxy to 29AB, regardless of the presence of emission lines, providing the first comprehensive test of photometric selection at these redshifts. The first galaxies could be largely continuum sources due to the attenuation of the Lyman-α line emission and extremely low metallicities. For galaxies with strong nebular emission lines at $z\sim9$, we can compile [OII], [NeIII], Hβ, and [OIII] to study star formation rates, metallicity, and excitation and the source of ionizing photons (e.g., AGN). Finally, we can directly confirm the presence of considerable Balmer breaks at z=9 (Hashimoto+2018), indicative of extremely early formation. Such a discovery would reset our picture of early galaxy formation, but would lend credence to the EDGES detection of 21cm (Bowman+2018) that suggests star formation must have been well underway by $z\sim20$. The evolution of the star formation density may even constrain the mass of the warm dark matter particle (Dayal+2018).

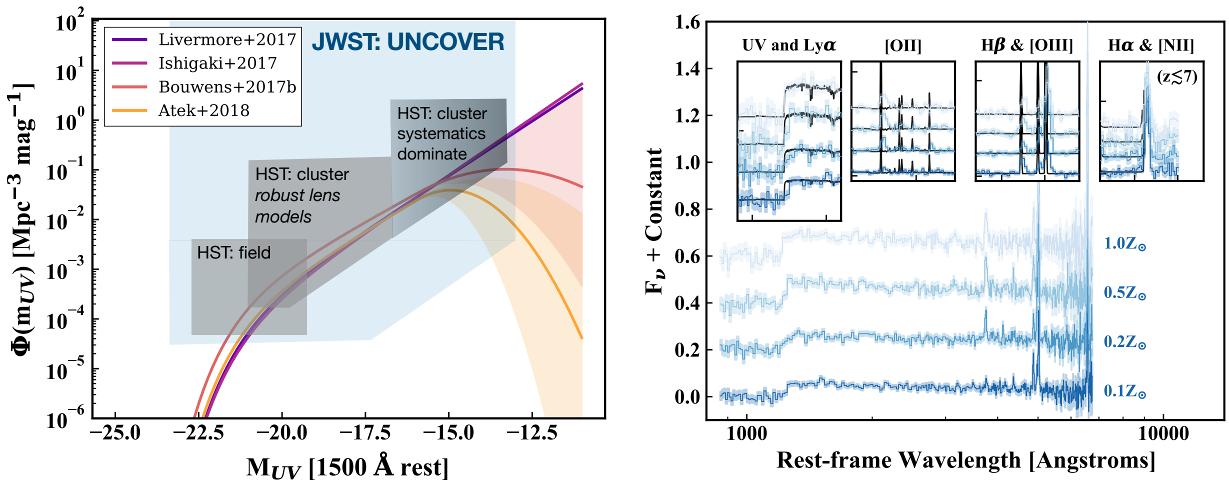

Figure 3: Left: Systematics at high magnification (>3 magnitude) dominate uncertainties at M_{UV}>-16, casting doubt on the faint-end turnover of the luminosity function at $z{\sim}6$ reported in some HST studies. UNCOVER will combine superior depths and spatial resolution to resolve this question. Right: Simulated NIRSpec/PRISM observations of $z{\sim}6.5$ faint AB=28.5 Lyman Break Galaxies. Hydrogen and Oxygen emission line ratios will constrain ionizing photon production rate and comparison of the UV spectral slope and high SNR measurements of Hydrogen lines may constrain the escape fraction at the epoch of reionization.

II. Reionization

Understanding how and when galaxies irradiated their environment and reionized the surrounding neutral gas is a key outstanding question. Measurements of CMB polarization place this epoch of reionization between z=6−10 (Planck Collaboration+2018). To determine the contribution of galaxies to reionization requires measuring: (1) the faintest galaxies that dominate the integrated UV light, (2) the rate at which they produce ionizing photons, and (3) the fraction of photons that escape. Until now, the latter two have been impossible to address due to the lack of λ>2.5μm spectra. Even studies of the UV LF have been mired in uncertainty. Recent studies of the Hubble Frontier Fields are discrepant at the faint end, with some showing a turn over at M_{UV}>-16 (Fig. 3, e.g, Bouwens+2017, Atek+2018) while others report a steep rise (e.g, Livermore+2017), dramatically changing the total light output of galaxies. The reason for the differences is that the shallower HST data required extreme magnification (> 20×) where the uncertainty in the lens models are large.

What UNCOVER enables: By reaching 1.5 magnitudes deeper than HST, measuring the turnover will no longer rely on highly magnified sources. In addition, the 3× spatial resolution of NIRCam versus HST/WFC3 further reduces the systematic uncertainties due to source size assumptions. UNCOVER probes securely to $M_{UV}{\sim}$-14 and should settle the issue of the potential turnover and directly measure 85% of the reionizing UV radiation.

Ultra-deep PRISM spectra will allow us to the study ionizing spectra of the faintest galaxies. Of the expected ∼1500 galaxies at z=6−7, we estimate about ∼40 will have intrinsic UV magnitudes of M_{UV}>-16 but still be bright enough (<28AB) to get high quality spectra. For

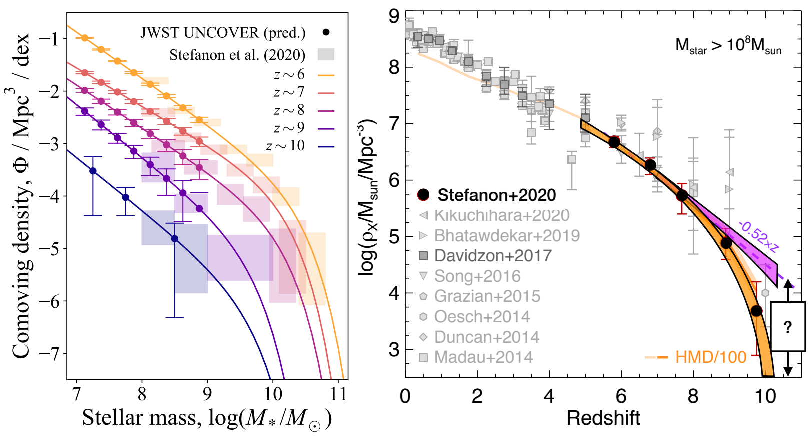

Figure 4: The stellar mass function is a key test for models of galaxy formation (e.g., Stefanon+2020). The evolution of stellar mass density is sensitive to the faint end slope, which requires ultra-deep data. The evolution at high redshift reflects the stellar mass-halo mass relation and probes the overall efficiency of galaxy formation.

these galaxies we observe the full complement of lines ([OII], [OIII], Hβ, Hα, [NII], [SII], with Hα and [NII] marginally resolved at $R\sim300$ at $\sim5\mu$m), to study ISM properties, ages, metallicity, and ionization mechanisms. Finally, Hydrogen line strengths relative to the UV slope may place constraints on the ionizing photon escape fraction, f_{esc} (e.g., Zackrisson+2017), with a 20hour PRISM spectra able to infer f_{esc}>0.5 to <28AB. The combination of imaging and spectra will lead to a quantum leap in studies of reionization era galaxies.

III. Legacy Science

The UNCOVER ultra-deep data enable Legacy science on a number of topics:

Stellar mass complete samples to z=10: Despite 10,000 HST orbits and tens of thousands of hours of Spitzer observations, our picture of the Universe remains highly incomplete due to reliance on HST-selection (rest-frame UV at z>4). This means we are biased towards the youngest, least dusty galaxies at these early epochs. From the deepest Spitzer data we know that there are many red galaxies that are missed entirely at HST wavelengths (e.g., Wang+2019). JWST will enable rest-frame optical (F444W) selections to extreme depths (~29 AB, 3 magnitudes deeper than Spitzer), enabling us to construct the first stellar mass complete samples to z=10. Furthermore, assisted by lensing (to ~32 AB equivalent) UNCOVER will allow us to reach $10^7 M_\odot$ at high SNR>5 for star forming galaxies to $z\sim10$. To calibrate our stellar masses we follow up a subset of galaxies with PRISM spectroscopy to confirm redshifts and determine the contribution of emission lines to photometry (the two main underlying uncertainties). The extreme depth of UNCOVER provides the best constraints on the faint end slope of the stellar mass function (see Fig. 4) and is the perfect complement to wider and shallower ERS/GTO data. Combined, the $7<z<10$ stellar mass function will allow unprecedented estimates of the evolution of the integrated stellar mass density to z=10 and, via abundance matching, on the evolution of the $M_\star$ to M_{halo} relation, which is critical for connecting observations of galaxies to theory and simulations.

Quenching: Quenching of star formation is a complex phenomenon and at different times, mass scales, and environments different processes are involved, which require both photometry and spectroscopy to address. Properties of quiescent galaxies place strong constraints

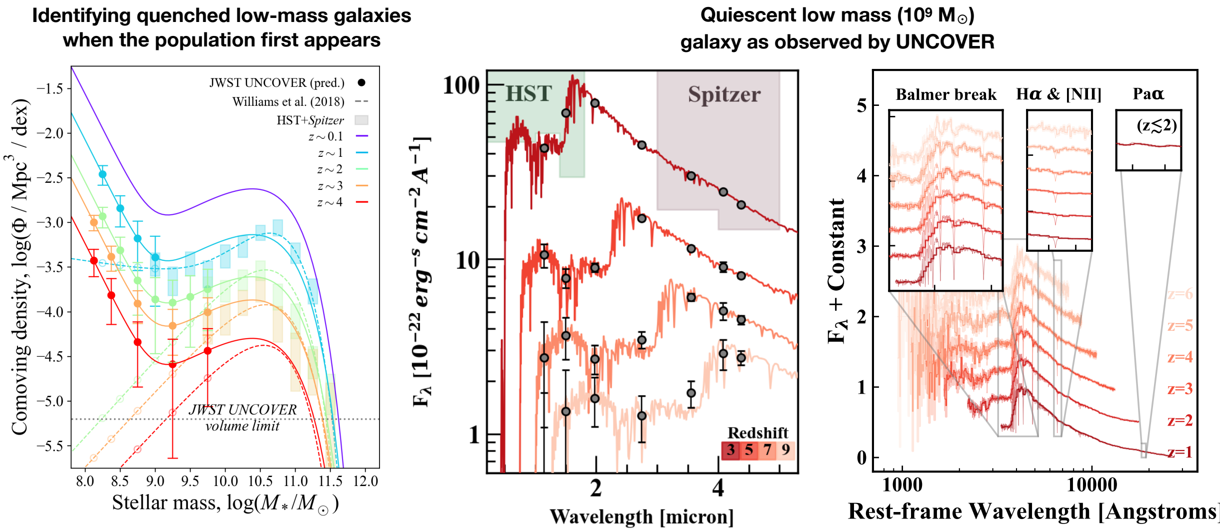

Figure 5: Left: Below $z\sim0.75$ the upturn in the number density of quiescent galaxies reflects satellite quenching. Although this signal seems to disappear at higher redshift, HST data are simply not deep enough. We know that dwarf ellipticals in the local Universe are older than the $\sim$7 Gyr lookback times, but have yet to identify younger progenitors. UNCOVER will identify and spectroscopically confirm quiescent satellites as early as the population emerges (center and right panels), thereby characterizing satellite quenching across cosmic time.

of feedback mechanisms in galaxy formation models. The earliest massive quiescent galaxies have now been confirmed to redshift $z\sim4$ (e.g., Forrest+2020) but this limit is entirely determined by our ability to detect them: quiescent galaxies almost certainly exist to lower masses and higher redshift beyond the capabilities of HST and ground-based spectroscopy.

UNCOVER will detect any quiescent galaxy in NIRCam to unprecedented low mass and high redshift, i.e., $<10^9 M_\odot$ to z=9 (see Fig. 5), and confirm the quiescence of select galaxies through the absence of emission lines and strong Balmer Breaks in PRISM spectroscopy. Beyond identification and counting, the PRISM spectroscopy is deep enough to determine stellar metallicities to $F444W<26$ and stellar ages to $z\sim7$ to $F444W<27.5$AB enabling independent constraints on scenarios for their quenching, and determining the duty cycle of low mass galaxies to $z\sim10$. UNCOVER will measure the onset of the satellite quenching upturn, thereby mapping quenching timescales of low-mass galaxies over $1<z<3$.

Star formation and dust: Dust is generally thought not to play a role in low mass galaxies at high-z (e.g., Bouwens+2015, Finkelstein+2015), but this is at least partially a selection effect as galaxies are rest-UV-selected. Highly obscured extreme starbursts are known to exist in shallow wide field sub-mm surveys; they represent only the tip of the iceberg. What lies in between is still terra incognita (see Fig. 6) and it remains unclear what fraction of star formation is still missed due to dust at $z>4$. ALMA is indeed finding dusty galaxies to much lower masses, star formation rates, and higher redshifts than expected (e.g., Williams+2019), but the drawback of ALMA is that it is inefficient for surveying (about 100$\times$ less area than

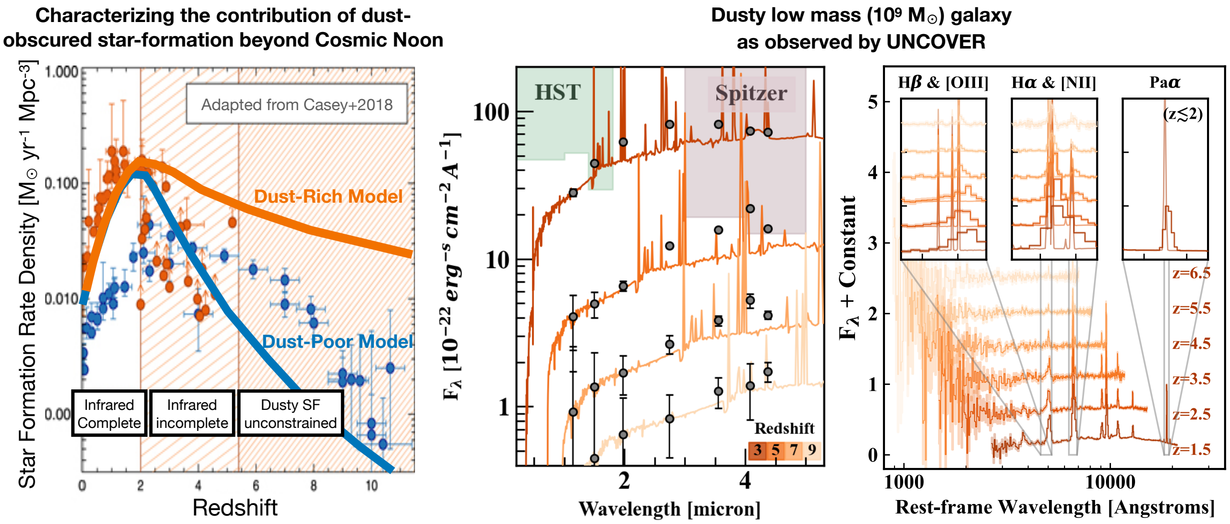

Figure 6: Left: Dust-obscured star formation is known to dominate the SFR density below Cosmic Noon ($z\sim2$), beyond which dust is not thought to play a large role in galaxies. However the census of dusty star formation is incomplete or unconstrained at higher redshift (Casey+2018). Center: predicted example photometry to our NIRCam limits will detect *any* obscured galaxy to $>10^9 M\odot$ to z=9, should they exist. Right: UNCOVER can spectroscopically confirm the existence of this population to $28AB$ mag.

NIRCam) and better suitable to follow up of galaxies found by JWST.

With the ERS/GTO capable of finding massive obscured galaxies over larger volumes, UNCOVER will instead put this question to rest at the lowest masses and the highest redshifts. The ability to detect any highly obscured $M_\star{>}10^9M_\odot$ galaxy to z=9 with NIRCam and test photometric stellar population inferences with NIRSpec spectroscopy (with Balmer decrement A_V), will finally reveal whether our census of star formation is significantly incomplete.

Further Science Opportunities with UNCOVER: • Strong line diagnostics, ionizing spectra, nebular continuum for low-luminosity AGN and star forming galaxies at $4<z<7$ • Rest-UV diagnostics may reveal AGN activity in low-mass galaxies, which could host intermediate-mass black holes/black hole seeds (Feltre+2016) • Known extreme targets in the field including: a galaxy overdensity at $z\sim8$ (Ishigaki+2016) and a multiply-imaged $z\sim10$ candidate (Zitrin+2014) • Highly magnified spatially resolved studies at $2-5\mu$m • An additional epoch and depth to the Windhorst GTO program • Synergistic NIRCam imaging over the ERS GLASS (PI: Treu) NIRSpec/NIRISS footprint.

UNCOVERing unknown unknowns: Perhaps the most exciting Legacy is the potential to discover objects we have not yet imagined. The advance in sensitivity, wavelength coverage, and spectral resolution is so large that we are almost guaranteed to run into surprises. **Covering ultra-deep parameter space with imaging and spectroscopy should be done early in JWST's mission** to ensure enough time for follow up in future cycles.